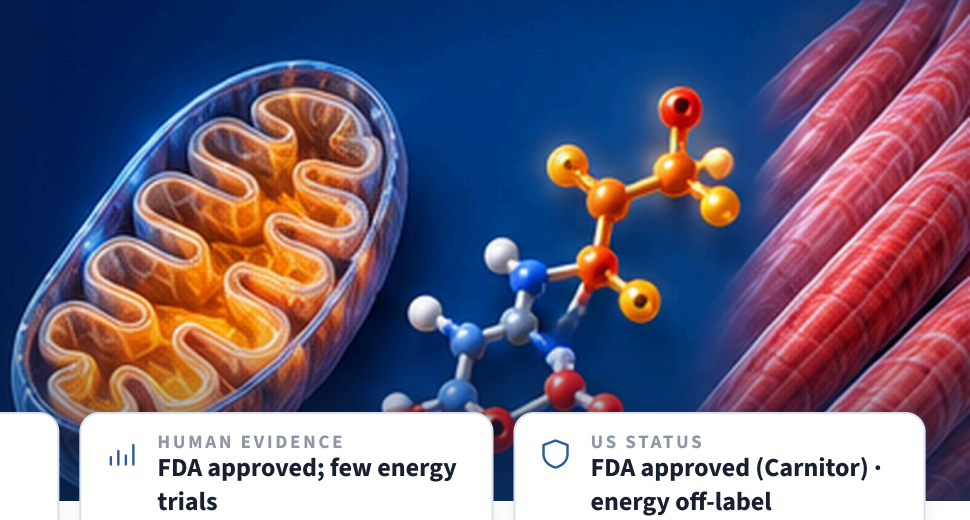
ENERGY 06

L-carnitine

Natural substance (levocarnitine) that moves long-chain fatty acids into mitochondria



CLASS
Nutrient (amino acid derivative)

STUDIED FOR
Carnitine deficiency, weight, fatigue

HUMAN EVIDENCE
FDA approved; few energy trials

US STATUS
FDA approved (Carnitor) · energy off-label

STUDIED FOR

- **Carnitine deficiency** — HUMAN
- **Body weight** — HUMAN
- **Fatigue on dialysis** — HUMAN
- **Weight during a fast** — HUMAN

Mostly oral supplement data. Injection studies found were small or in dialysis patients, not healthy people seeking energy or fat loss.

HOW IT WORKS IN THE BODY

L-carnitine ▶ Fatty acids ▶ Mitochondria ▶ Energy use

- **Label:** moves long-chain fatty acids into mitochondria, where they are burned
- **Required** in mammalian energy metabolism, the Carnitor label says
- **In dialysis,** effects on symptoms and outcomes “have not been determined” (label)

EVIDENCE & WHAT’S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- FDA approved (Carnitor) for carnitine deficiency
- Oral meta-analysis: 37 trials, 2,292 people
- No study of compounded carnitine blends

NOT YET ESTABLISHED

- Injections for energy or fat
- Effects in healthy men
- Compounded blends (not approved)

QUICK PRACTICAL INFO

HALF-LIFE
Terminal half-life 17.4 hours (IV, label)

ROUTES STUDIED
By mouth (tablets, solution); IV injection (label)

PHARMACOKINETICS
Oral bioavailability about 15%; about 76% of IV dose in urine in 24 h

SOLD AS
Prescription (Carnitor, generics); compounded blends not approved

MONITORED IN TRIALS
Plasma carnitine, blood chemistries, vital signs; INR on warfarin

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved (Carnitor) for carnitine deficiency
FDA	Compounded carnitine blends: not approved
WADA	Not listed; IV over 100 mL per 12 h banned (M2.2)

REFERENCES

1. FDA (DailyMed), Carnitor (levocarnitine) Injection label, Nov 2023
2. Clin Nutr ESPEN, L-carnitine and body weight meta-analysis, Jun 2020
3. WADA, 2026 Prohibited List

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